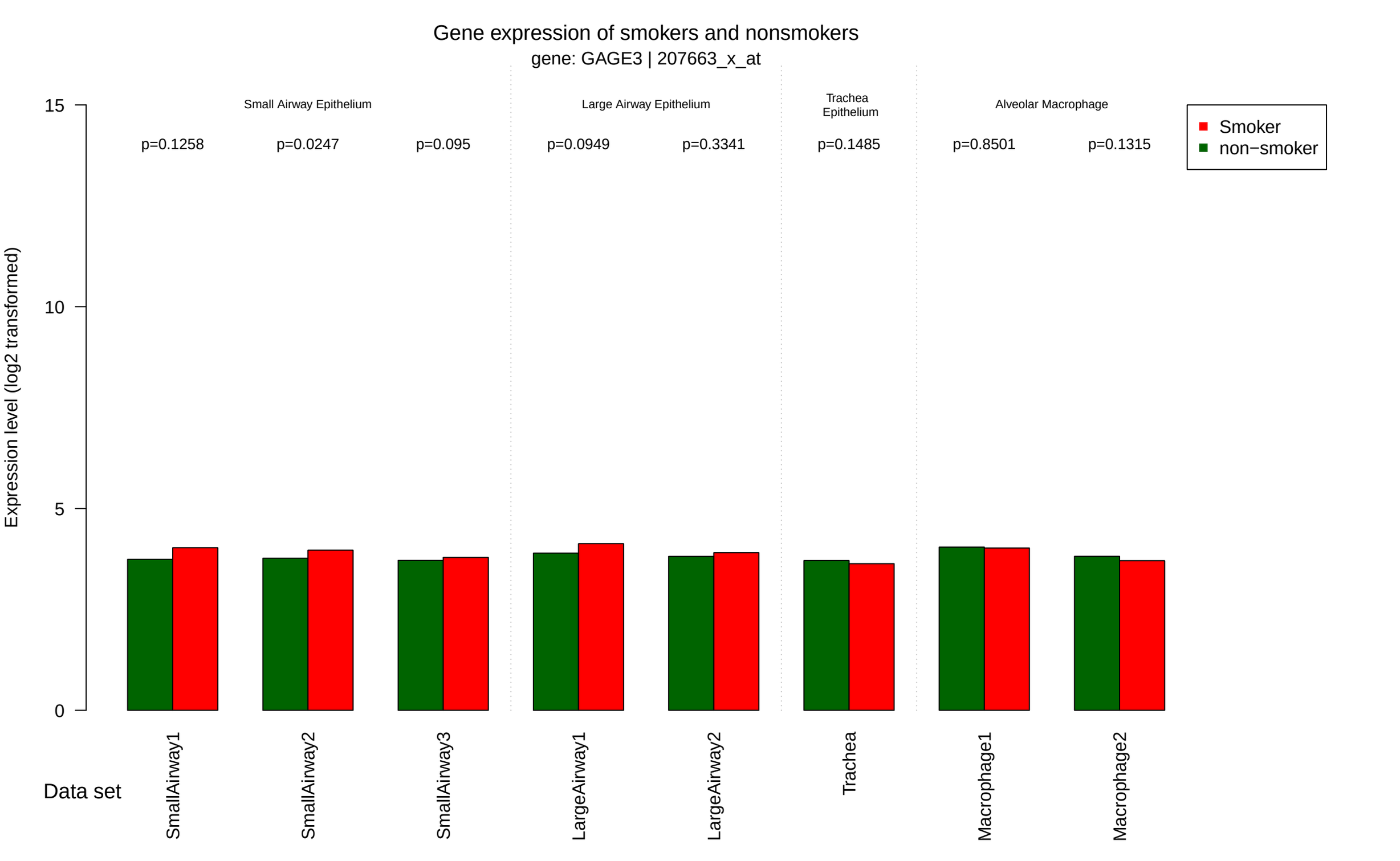
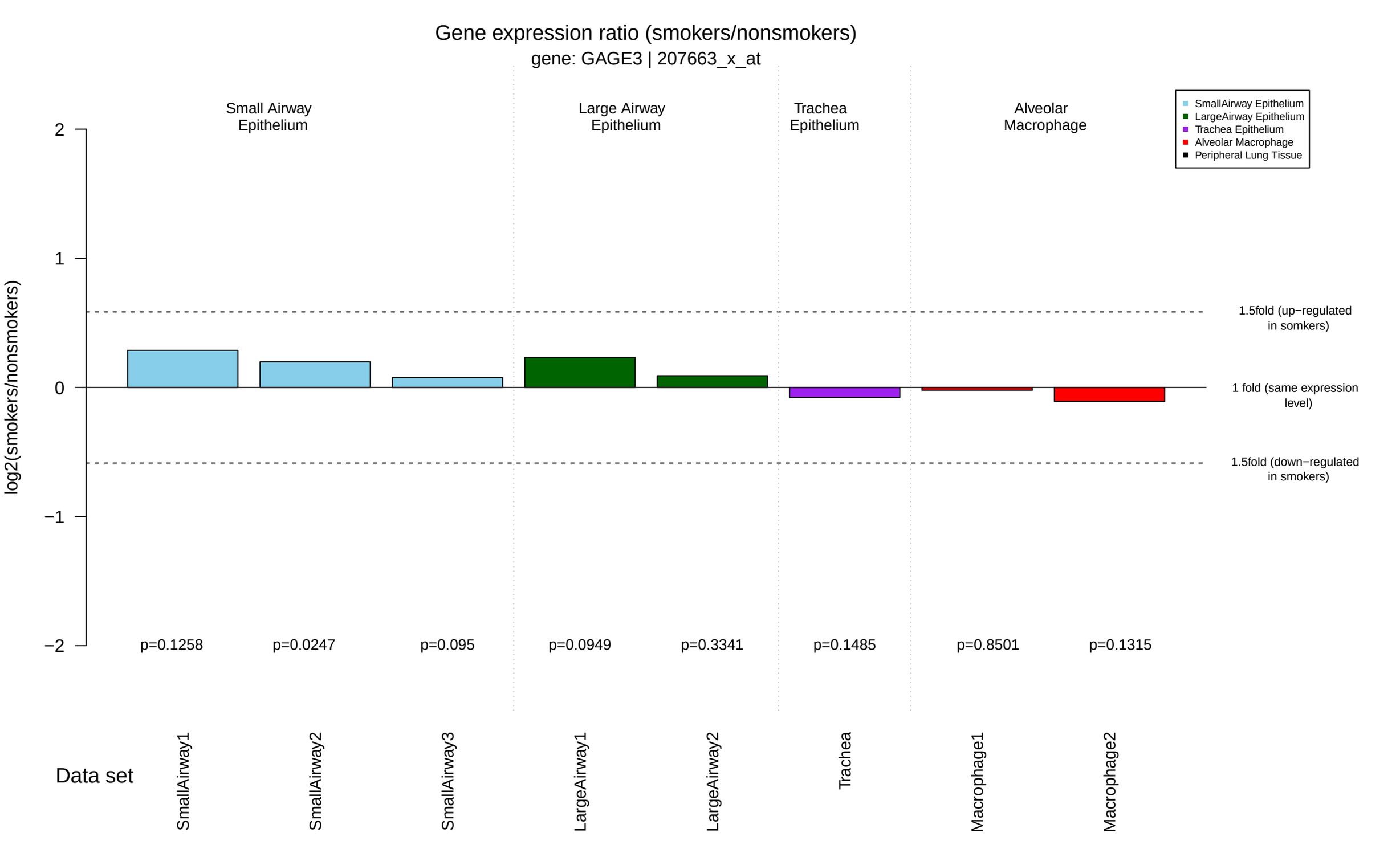
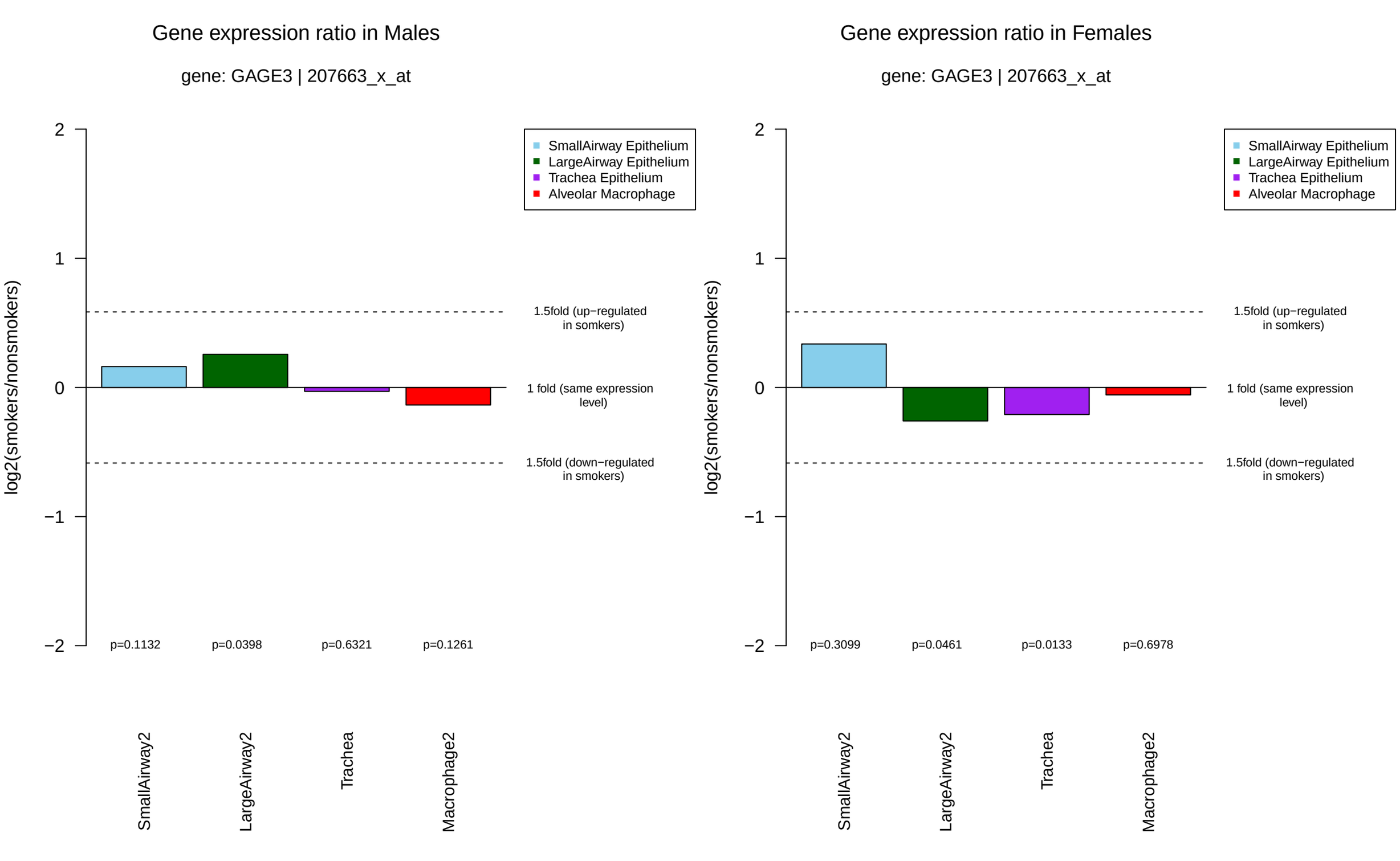
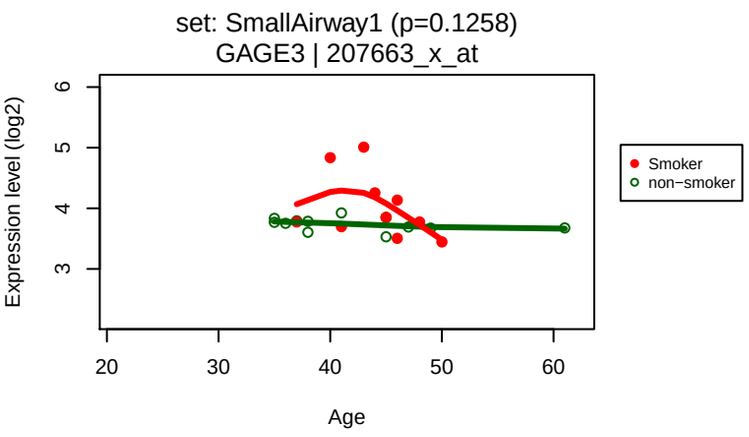
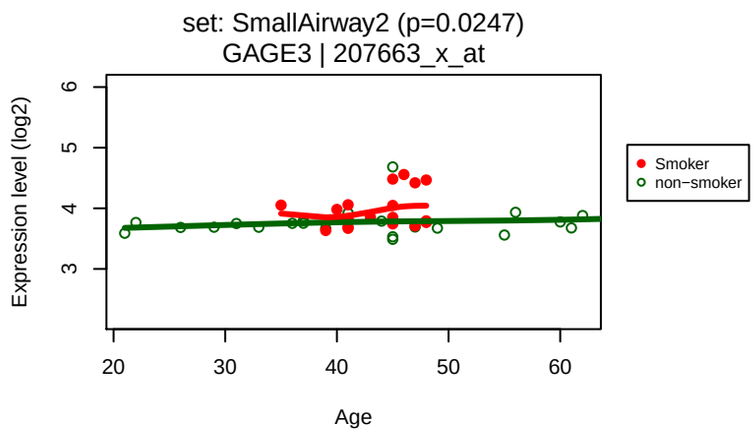
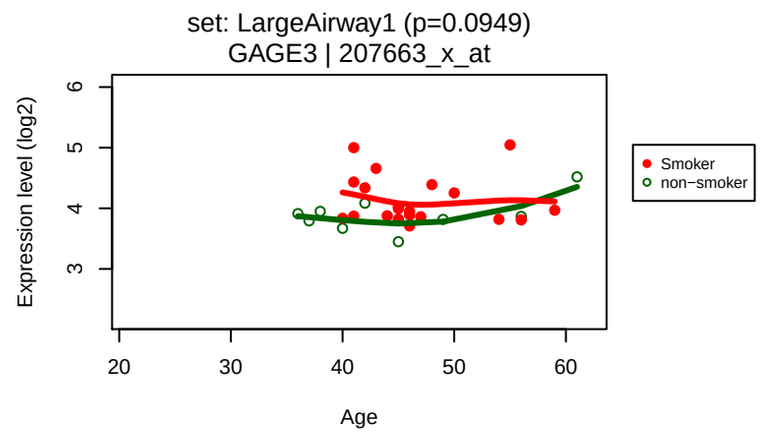

Gene expression of smokers and nonsmokers
gene: GAGE3 | 207663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1258
p=0.0247
p=0.095
p=0.0949
p=0.3341
p=0.1485
p=0.8501
p=0.1315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

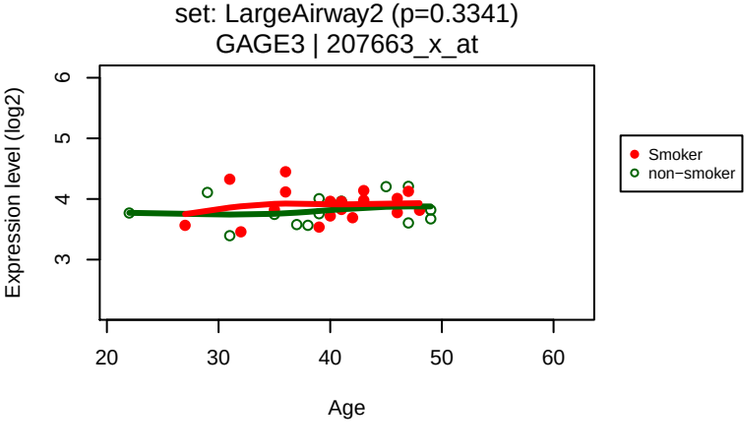

Gene expression ratio (smokers/nonsmokers)
gene: GAGE3 | 207663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1258
p=0.0247
p=0.095
p=0.0949
p=0.3341
p=0.1485
p=0.8501
p=0.1315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

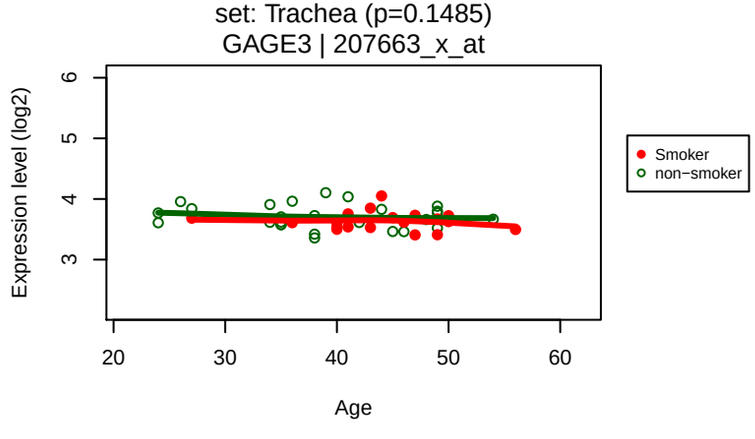

Gene expression ratio in Males
gene: GAGE3 | 207663_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1132
p=0.0398
p=0.6321
p=0.1261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAGE3 | 207663_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3099
p=0.0461
p=0.0133
p=0.6978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

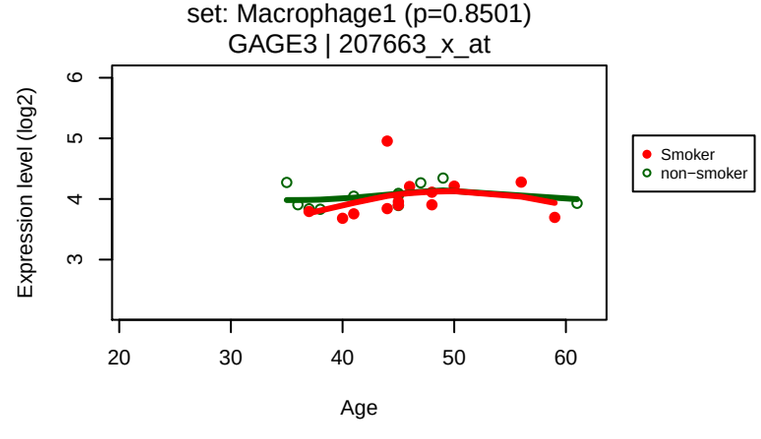

set: SmallAirway1 (p=0.1258)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker

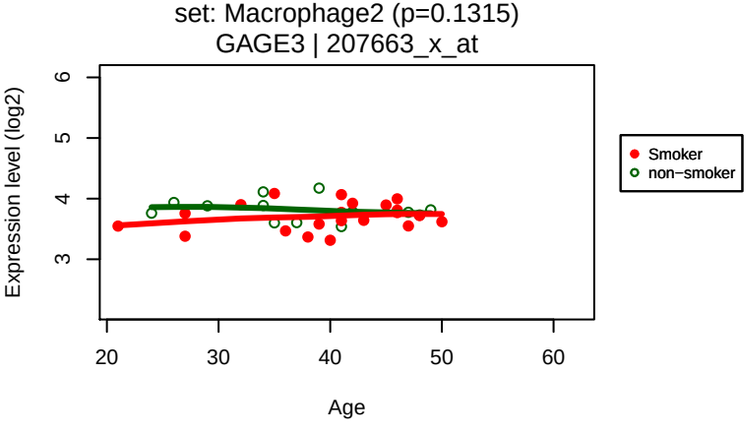

set: SmallAirway2 (p=0.0247)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker

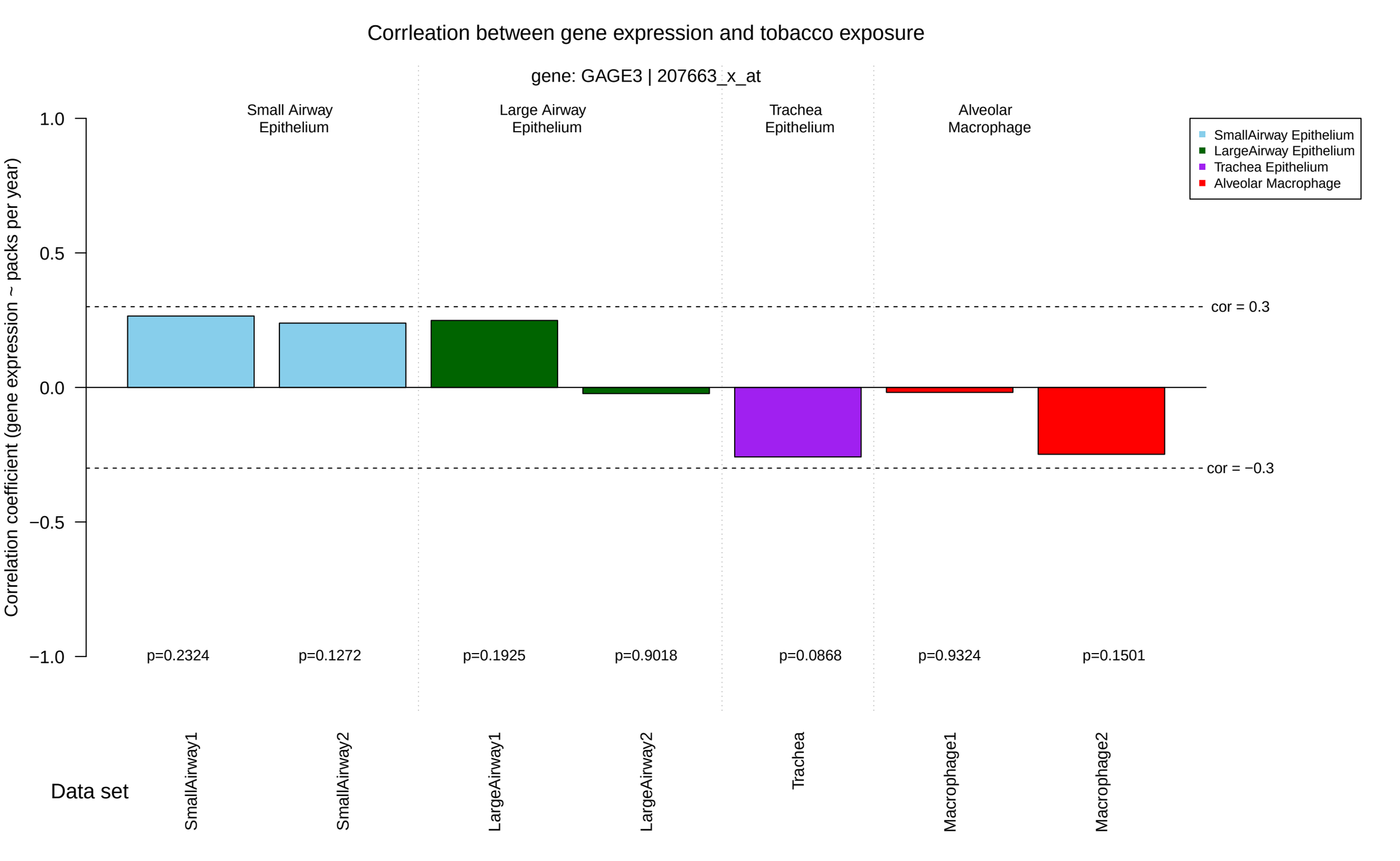

set: LargeAirway1 (p=0.0949)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3341)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1485)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8501)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1315)
GAGE3 | 207663_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAGE3 | 207663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2324
p=0.1272
p=0.1925
p=0.9018
p=0.0868
p=0.9324
p=0.1501
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2